

A systems approach

Instruction is a systematic process in which every component (i.e. teachers, students, materials, and learning environment) is crucial to successfully learning (Dick & Carey, 1996). Instruction deals with teaching and learning activities. These activities should assist students to learn knowledge and move this knowledge from short term memory to long term memory. To do that, students need to learn how to rehearse, encode, process and feedback new knowledge to be able to remember when they need.

Educational Objectives ", the process of course or curriculum design are represented schematically

- Formulate objectives
- Select appropriate instructional methods
- Operate course or curriculum
- Access and evaluate

The new instructional design model is based on active learning. During teaching and learning activities, learner is active and uses cognitive learning to construct new knowledge. To construct new knowledge, educational technology materials are used. These materials are related with goals and objectives.

This process can be used to plan a variety of instructional approaches, ranging from teacher lectures to hands-on student-centered activities. In addition, as a result of using this process, teachers should be able to develop effective instruction. This effective instruction can help students to learn more and keep the new knowledge into long term memory. These students will be motivated to join class activities.

New model is described a five-step systematic planning process. These are:

1. Input
2. Process
3. Output
4. Feedback
5. Learning

Input step in the instructional planning because gives teacher information about the effectiveness of the instruction. In other words, these steps can help instructor to identify what to teach and how to teach instructional activities. The input step has five stages. These are:

1. Identify needs,
2. Identify contents,
3. Identify Goals-Objectives,

4. Identify teaching methods,
5. Identify instructional media.

The **process** step has three stages. These are test prototypes, redesigning of instruction and teaching activities. The first is to test prototypes. In this step, teacher will be ready to try out the planned instruction with the students. The main goal of first stage is to find out which stages are working and which stages are not working. In other words, the problems in instructional design are identified during testing prototypes. Testing prototypes tells instructor what students really want to learn and how to get there. The second stage is to redesigning of instruction. After problems are identified, instructional designer reorganizes instructional activities. To reorganize instructional activities, pre-testing plays a key role to design an effective instruction. If an effective instruction is designed well, instructional goals will be achieved successfully. The last stage is to teaching activities. Teacher begins teaching activities in terms of content, teaching methods, goals and objectives with instructional media.

The **output** step contains two stages. These are assessment and revising instruction. In the first stage, teacher assists teaching and learning activities in the instructional design model. Instructional designer uses formative and summative evaluation methods to check goals and objectives. This process requires teacher to implement assessment tools to determine whether the students did demonstrate the skills, knowledge, and attitudes that teacher described in instruction goals and objectives or not. When the students participate in the instructional activities, teachers want to know whether they learned what the instructional plan expected them to learn. To determine student learning, educational measurement and evaluation process should be implemented by teachers. This process gives teachers results on what students learn from the instruction. Teachers should analyze the results and make decision on where to go in the instruction. In the last stage, instructional designer evaluate all instructional activities. Instructional designer finds problems during the instructional design process. Then, instructional designer solve the problems then redesign the instruction.

The **feedback** step has one stage. This is “Go back to related step”. The feedback process involves revise instruction based upon the data collected during the implementation phase. If, during the phase, teacher finds that students are not learning what the plan wanted them to learn, and/or they are not enjoying the learning process, teacher will want to go back to related step and try to revise some aspect of their instruction so as to better enable their students to accomplish their goals. If there is a problem in input step, instructional designer will go back to input step. Then, instructional designer will make changes and start process from input. This process will be done until all goals and objectives are learned by learners. During this cycle, instructional designer may go back to any steps to where a problem is occurred.

The **learning** step has one stage. This is “Long Term Learning”. The learning process involves full learning. In this process, teacher wants to make sure that their students have learned what the instructional plan wanted them to learn. If, during the phase, teacher finds that their students accomplished their goals in the instructional activities, teacher will want to go new instructional activities. At the end of this step, long term learning is accomplished by instructional designer.

REFERENCES

- Best, J.W. & Kahn, J.V. (1993). Research in Education. Allyn and Bacon, Boston.
- Duffy, T.M. & Jonassen, D.H. (1992). Constructivism and the Technology of Instruction. Lawrence, New Jersey USA.
- Dick, Walter & Carey, Lou. (1996). The Systematic Design of Instruction. Fourth Edition. Harper Collins College Publishers, New York USA.
- Gagne, R.M.; Briggs, L.J.; Wager, W.W.; Golas, Katharine C.; & Keller, John M. (2005). Principles of Instructional Design. Wadsworth USA.
- Gustafson, K.L. (1996). International Encyclopedia of Educational Technology. Edited by Plomp, T. & Ely, A.P. Pergamon, USA.
- Kemp, J.E.; Morrison, G.R. & Ross, S.M. (1994). Designing Effective Instruction. Merrill, New York USA.
- McGriff, J.S. (2001). ISD Knowledge Base / Behaviorism. URL <http://www.personel.psu.edu>. (Retrieved on October 10 2004).
- McGriff, J.S. (2001). ISD Knowledge Base / Constructivism. URL <http://www.personel.psu.edu>. (Retrieved on October 10 2004).
- McGriff, J.S. (2001). ISD Knowledge Base / Cognitivism. URL <http://www.personel.psu.edu>. (Retrieved on October 10 2004).
- Reiser, Robert A. & Dick, Walter. (1996). Instructional Planning: A guide for teachers. Allyn and Bacon, Boston USA.
- Seels, Barbara B. (1995). Instructional Design Fundamentals. Educational Technology Publications, Englewood Cliffs, New Jersey USA.

LEWIS ELTON MODEL OF INSTRUCTIONAL TECHNOLOGY

There are three categories

- Mass instruction techniques,
- Individualized learning techniques and
- Group learning techniques

Three basic phases of Instructional methods

Mass Instruction

Conventional lectures and expository lessons; television and radiobroadcasts; cable television; films and videos are included in mass instruction.

Mass instruction is, of course, as old as education itself, with the 'lecture' and 'expository lesson' being the dominant instructional techniques in virtually all sectors of education and training.

throughout history. It was, however, only in the period following the Second World War that a systematic effort was made to improve the efficiency and cost-effectiveness of the method by using the new types of hardware that were starting to become available. By such means, it was hoped that more people could be educated or trained without necessarily increasing the number of teachers or trainers, and that the overall effectiveness of the teaching process could be improved. Some important outcomes were the development of basic mass instruction tools like the overhead projector and 35mm slide projector, and the increasingly wide spread use of 'hardware-based' technique such as film, radio and television broadcasting and closed-circuit television.

Role of teachers

Traditional expository role; controller of instruction process.

Techniques

Lectures and similar expository techniques.

Strengths

- Can be very cost effective in terms of student/staff ratio.
- Strong in achieving lower cognitive and some affective objectives.
- Generally popular with both students and staff.

Film and video presentations.

Strengths

- Can be a highly-effective substitute for a lecture or part thereof if the content and level are suitable.
- Can be used to provide realistic illustrative, supportive, background and case-study material.
- Tend to be highly stimulating.

Educational broadcasts.

Strengths

Same basic strengths as film and video presentations, with further advantage that broadcasts are free.

Mass practical and studio work.

Strengths

- Can be effective in developing psychomotor and associated skills.
- Can help demonstrate relevance of theoretical content of a course.
- Students generally enjoy their participative nature.

2. Individualized Learning

Programmed learning; mediated self-instruction; computer-based learning are included in individualized learning.

Although individualized learning, in the form of correspondence courses and similar systems, also has a long tradition of use in education, it was only comparatively recently that it became part of main-stream educational technology. The catalyst for this development was behavioral psychology, which was pioneered by B F Skinner and his followers during the 1950's. Skinner's work on the stimulus/response mechanism, which represented (in many people's view) the first truly 'scientific' theory of learning, first triggered off the bandwagon programmed learning movement that dominated educational thinking during the 1960's. More recently, it led to the development of a wide range of individualized learning techniques (such as tape-slide and the various computer-assisted systems that are now achieving more and more widespread use) as well as to fully-integrated individualized instruction systems such as the Keller Plan and Open Learning systems.

Role of teachers

Producer/manager of learning resources; tutor and guide.

Techniques

Directed study of material in textbooks.

Strengths

- Can be a highly effective way of teaching basic facts, principles, applications, etc., provided that suitable texts are available and the work is carefully structured.
- Allows learner to work at his/her own natural pace.
- Needs no specialized facilities other than suitable library.

Study of specially-prepared notes or programmed texts.

Strengths

- Same basic advantages as directed study of books, and can be even more effective if the material is well prepared.
- Can allow learners to interact with the material.

Self-instruction via audiovisual media (audiotapes, video-tapes, tape/slide programmes etc.)

Strengths

- Enables a wide range of educational objectives to be achieved (especially 1wer cognitive).
- Allows learner to work at his/her own pace.
- In addition, use of mediated presentation enables sound, movement, realism, etc. to be introduced, thus increasing student stimulation.

- Can save teachers from having to carry out repetitive, time-consuming work (e.g. teaching certain basic laboratory skills).

Computer-based learning.

Strengths

- Enables a wide range of educational objectives to be achieved (especially lower cognitive).
- Allows learner to work at his/her own pace.
- Can allow considerable Inter-action between learner and Instructional programme, and can adapt to needs of learner; can be highly stimulating.
- Can provide (through computer simulations) a wide range of otherwise inaccessible learning experiences.
- Allows on-going assessment and monitoring to take place automatically.

Individual practical studio or project work

Strengths

- Same basic strengths as mass practical and studio work.
- Allows students to work at their own pace.

3. Group Learning

Tutorials; seminars; group projects; games and simulations are included in group learning.

While it can be argued that the 'individualized learning' phase of educational technology probably had a greater impact on modern education and training than the 'mass instruction' phase that preceded it, there are, in practice, a number of limitations to the approach. One of the most obvious stems from the fact that it is, by definition, individual, and, as such, cannot enable students to interact with one another and develop group skills such as discussion skills, interpersonal skills and the various other skills needed to collaborate effectively with other people in carrying out a common task or project. This has led to an increasing realization that the various activities that come under the general heading of group learning have a very important role to play in modern education and training.

Group learning installing the final stages of its 'development' phase, although its 'use' phase is now well underway - as evidenced by the widespread use now being made of participative exercises such as games and simulations.

Role of teachers

Organizer and facilitator.

Techniques

Buzz sessions and similar short small-group exercises.

Strengths

- Constitute an excellent method of Introducing variety into lecture, thus helping to maintain student attention.
- Can be used to achieve a wide range of objectives, both cognitive and non-cognitive.
- They get students actively Involved In a lesson.
- They allow feedback to take place.

Class discussions, seminars, tutorials, etc.

Strengths

- Same basic advantages as buzz sessions, etc. In addition, their greater length allows an even wider range of objectives to be achieved, often of a very high level.
- Enable relevant topics to be examined in great depth.

Participative exercises of the game/simulation/case study type.

- They can be used to achieve a wide range of objectives, both cognitive and non-cognitive often of a very high level.
- A High student involvement.
- Highly stimulating and motivating If proper) designed.
- Ideal for cross - disciplinary work.

Mediated feedback sessions such a micro teaching, recorded interviews, or recorded group exercises.

Strengths

- Use of mediated feedback (eg audio or video recording) enables valuable group discussion of student performance to take place.
- Can be used to develop a wide range of skills.
- High student involvement.

Group projects

Strengths

- Suitable for developing a wide range of objectives, both cognitive and non-cognitive, often at a very high level.
- Ideal for developing interpersonal and group skills.
- Ideal for cross-disciplinary work.

Further reading

1. A Handbook of Educational Technology, by F Percival and H I Ellington: Kogan Page, London; 1984. (An introductory text on educational technology specially written for practicing teachers and lecturers; it provides detailed guidance on how to select suitable instructional methods for different purposes).
2. Designing Instructional Systems, by A .1 Romiszowski; Kogan Page, London; 1981. (The definitive text on the subject, but probably a bit high-powered for the average teacher or lecturer - or the average educational technologist,